

PROJECT DEFINITION DOCUMENT (PDD)

The "Source of Truth"

1.1 BUSINESS OBJECTIVES

Problem Statement:

St. Michael's Hospital lacks visibility into billing discrepancies, patient visit patterns, and departmental performance. Revenue leakage occurs through undetected insurance miscalculations, while operational inefficiencies from no-show appointments and low patient satisfaction remain unquantified.

KPIs & Success Metrics:

KPI	Target	Measurement Method
Billing Accuracy	100% reconciliation	Calculated Patient Due vs. Total Paid variance =
Revenue Leakage Identification	100% of discrepancies flagged	# of rows where formula detects under/overpayment
Insurance Coverage Validation	0 errors in coverage %	Verify InsuranceCoveragePct against policy rule
Patient Satisfaction Tracking	Department average > 4.0	AVERAGEIFS by department
Readmission Rate Monitoring	Flag all 30-day returns	COUNTIFS per patient ID within date range
Data Quality Score	>90% across all records	Average of DataQualityScore column

QUALITY ASSURANCE (QA) REPORT

Data Cleaning Validation:

Issue Type	Count	Resolution Method	Status
Mixed Date Formats	100%	TEXT(...,"MM/DD/YYYY") standardization	✓
Medical Abbreviations	18 terms	SUBSTITUTE lookup chain	✓
Excel Serial Dates	40+ rows	Manual conversion verified	✓
Duplicate Department Names	3 variants	PROPER() standardization	✓
Missing Cancellation Reasons	28 rows	"Not Specified" default	✓
Insurance Coverage % as text	12 rows	Converted to number/100	✓
Doctor ID mismatch	0	VLOOKUP validated	✓
ICD-10 code formatting	100%	Standardized	✓

Deliverable: A self-contained Excel workbook that ingests raw patient visit data, performs automated cleaning, calculates financial reconciliation, and outputs operational dashboards for management review.